



Wrocław
University
of Science
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Car Rental Management System

Course: Software System Development, Project, group 2

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2. Requirements specification (**S2**)

2.1. Functional Requirements Specification

Symbol	Type	Description	Significance	Source
FR-01	UI / Business Logic	Customers should be able to filter and sort vehicles by type, make, model, drive type, and price.	MUST	Customer (CUST)
FR-02	Business Logic	The system must calculate the total rental cost and present the rental conditions before confirmation.	MUST	Customer (CUST)
FR-03	Data Exchange	The system must initiate and process payment transactions through an external payment provider.	MUST	Payment Provider (PAY)
FR-04	Business Logic	The system must automatically generate a rental agreement upon successful payment confirmation.	MUST	Gov. Authorities (GOV) / Customer
FR-05	Reporting	The system should allow the generation of reports on rental activity, revenue, and fleet utilization for any selected time period.	SHOULD	Employee (EMP)

SSD 2026 - Project Report **S2**

FR-06	Data Exchange	The system may track the vehicle location during the rental period using GPS integration.	COULD	GPS Tracking System
FR-07	UI / Business Logic	The system must allow users to browse available vehicles	MUST	Customer (CUST)
FR-08	UI / Business Logic	The system must display detailed information about each vehicle	MUST	Customer (CUST)
FR-09	Business Logic	The system must allow users to create a vehicle reservation	MUST	Customer (CUST)
FR-10	UI	The system must allow users to enter personal data for a reservation	MUST	Customer (CUST)
FR-11	UI / Business Logic	The system must allow users to register an account	MUST	Customer (CUST)
FR-12	UI / Business Logic	The system must allow users to log in and log out	MUST	Customer (CUST)
FR-13	Business Logic	The system must allow customers to cancel an existing reservation	MUST	Customer (CUST)

SSD 2026 - Project Report S2

FR-14	Data Exchange	The system must allow customers to download their rental agreement	MUST	Customer (CUST)
FR-15	Business Logic	The system must allow employees to process vehicle pickup, including reservation verification	MUST	Employee (EMP)
FR-16	Business Logic	The system must allow employees to process vehicle return, including condition inspection and damage recording	MUST	Employee (EMP)
FR-17	Business Logic	The system must allow employees to add, update, and remove vehicles from the fleet	MUST	Employee (EMP)
FR-18	Business Logic	The system must allow employees to manage customer data	SHOULD	Employee (EMP)
FR-19	Business Logic	The system must allow employees to manage reservations	MUST	Employee (EMP)
FR-20	UI	The system must allow customers to view their rental history	SHOULD	Customer (CUST)

2.2. Non-Functional Requirements

Symbol	Type	Description	Significance	Source	Verification method
NF-01	Performance	The system must allow completing a reservation in under 5 minutes	MUST	Customer (CUST)	Performance testing
NF-02	Performance	The system response time will not exceed 2 seconds	SHOULD	Customer (CUST)	Load testing
NF-03	Performance	The system must support multiple concurrent users	MUST	Internal (EMP)	Stress testing
NF-04	Usability	The system must be accessible via modern web browsers	MUST	Customer (CUST)	Compatibility testing
NF-05	Usability	The reservation process should be simple and require minimal steps	SHOULD	Customer (CUST)	Usability testing
NF-06	Scalability	The system must support cloud and on-premise deployment	MUST	Internal (EMP)	Deployment testing
NF-07	Scalability	The system must scale with increasing users and reservations	MUST	Internal (EMP)	Scalability testing
NF-08	Security	The system must comply with GDPR	MUST	Gov. Authorities (GOV)	Security audit

SSD 2026 - Project Report **S2**

NF-9	Security	The system must ensure the secure storage of user data	MUST	Gov. Authorities (GOV)	Penetration testing
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2.3. Use Case Diagram

